

## Molecular Tumor Board Recommendation May 9, 2025

Dear Dr. Patel:

The Markey Cancer Center Molecular Tumor Board met on May 6, 2025 and reviewed your patient, Darrell Mills (110834266; non-small cell lung carcinoma with choriocarcinomatous differentiation). After reviewing the genomic report (Caris report dated April 7, 2025) and clinical characteristics of the case, the MTB recommends consideration of the following:

1. If progression, consider germ-cell based therapy regimen including carbo-etoposide with pembrolizumab targeting the high tumor mutation burden of 29. This is a category 1 evidence level recommendation.
2. If progression, consider on-label trastuzumab-deruxtecan due to the HER2 3+ result. This is a category 1 evidence level recommendation.
3. Please note, when we review the morphology of the pathology slides, the immunohistochemical protein profile, and molecular data, the findings in this case are consistent with a metastatic non-small cell lung carcinoma with choriocarcinomatous-like differentiation. Rare reports of primary lung choriocarcinoma have been reported. Those previously reported tumors are likely synonymous with this case, or at least represent one end of the spectrum of high grade carcinomas showing trophoblastic-like reprogramming.

Thank-you for submitting this case to the Markey Cancer Center MTB.

Signature:  Digitally signed by Susanne Arnold  
Date: 2025.05.14 12:19:48 -04'00' Date: \_\_\_\_\_

Signature: **Justine Pickarski** Digitally signed by Justine Pickarski  
Date: 2025.05.12 10:55:26 -04'00' Date: \_\_\_\_\_

Signature: **Rachel W. Miller, MD** Digitally signed by Rachel W. Miller, MD  
Date: 2025.05.13 16:48:13 -04'00' Date: \_\_\_\_\_

| Level of Evidence | Interpretation                                                                                                                           | Supporting Evidence                                                                     |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 1                 | FDA-labeled indication or NCCN-recommended; substantial evidence supporting use (standard of care)                                       | Meta-analysis, RCTs, etc. OncoKB level 1                                                |
| 2                 | Off-label use; substantial evidence supporting use                                                                                       | Phase II studies and above OncoKB level 2 and above                                     |
| 3                 | Off-label use; low to moderate evidence supporting use; MTB recommends, clinical trial preferred                                         | Phase I, case studies and reports OncoKB level 3 and below                              |
| 4                 | Off-label; no evidence supporting use; enrollment in clinical trial recommended; report experience to MTB if used outside clinical trial | Little to no evidence (pre-clinical, animal/cellular models), mechanistically plausible |